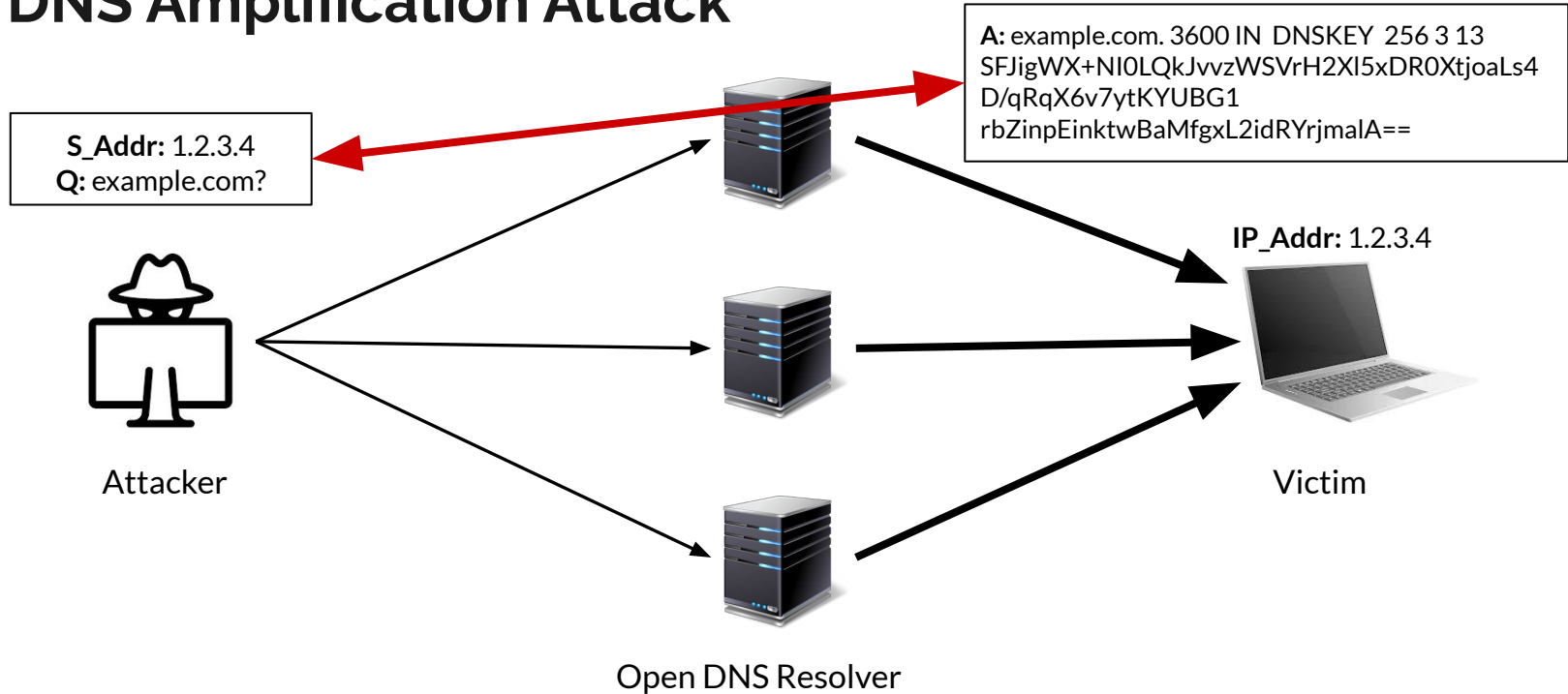




# **An Assessment of DNSSEC Amplification in 2026**

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# DNS Amplification Attack





# DNSSEC

- DNS Security Extensions introduced by IETF in 2005
- Address fundamental insecurity of DNS
  - Adds integrity and authenticity
  - Domain operators sign their zones, allowing everyone who queries DNS to validate the signatures
- Adds record types to DNS → **Does this increase the risk of amplification?**



## DNSSEC Record Types

- **DNSKEY:** Contains the public keys used to sign a zone
  - Zone Signing Key (ZSK) & Key Signing Key (KSK)
- **DS:** Contains a hash of the child zone's KSK
- **RRSIG:** Contains the DNSSEC signature for a record set
  - Every signed RRset in a zone has a corresponding RRSIG, so zones with many record types produce many RRSIG records
- **NSEC:** Links to the next record name in the zone to prove non existence of a record
  - NSEC3 uses hashed versions of names



# Research Questions

## DNSSEC Signed vs. Unsigned Domains

- RQ1: What is the distribution of signed & unsigned domains today?

## Amplification Behavior

- RQ2: What is the distribution of amplification factors for DNSSEC enabled queries?
- RQ3: Which DNSSEC enabled queries produce the highest amplification?

## Causes of Amplification

- RQ4: How do amplification factors differ between signed vs unsigned domains?
- RQ5: How do DNSSEC properties (i.e., key size, algorithm) affect amplification?

# Methodology

- **Dataset:** Tranco top 1M
- **Queries:** DNSKEY, DS, RRSIG, NSEC
- We calculate amplification as:

$$\text{Amplification Factor} = \frac{\text{Response size (bytes)}}{\text{Query size (bytes)}}$$

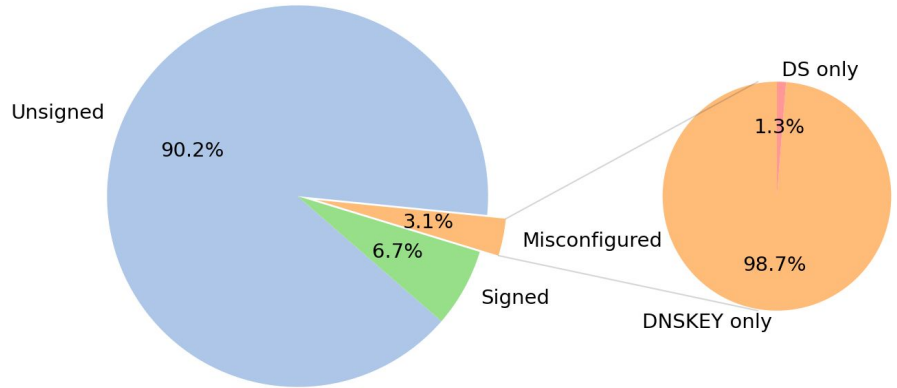
```
dig @8.8.8.8 example.com QUERY  
+dnssec +bufsize=4096 +norecurse  
+qr +stats
```

```
ANSWER example.com. 3600 IN  
DNSKEY 256 3 13  
SFJigWX+NI0LQkJvvzWSVrH2Xl5xDR0Xtj  
oaLs4D/qRqX6v7ytKYUBG1  
rbZinpEinktwBaMfgxL2idRYrjma1A==
```

## RQ1: DNSSEC Deployment

We define:

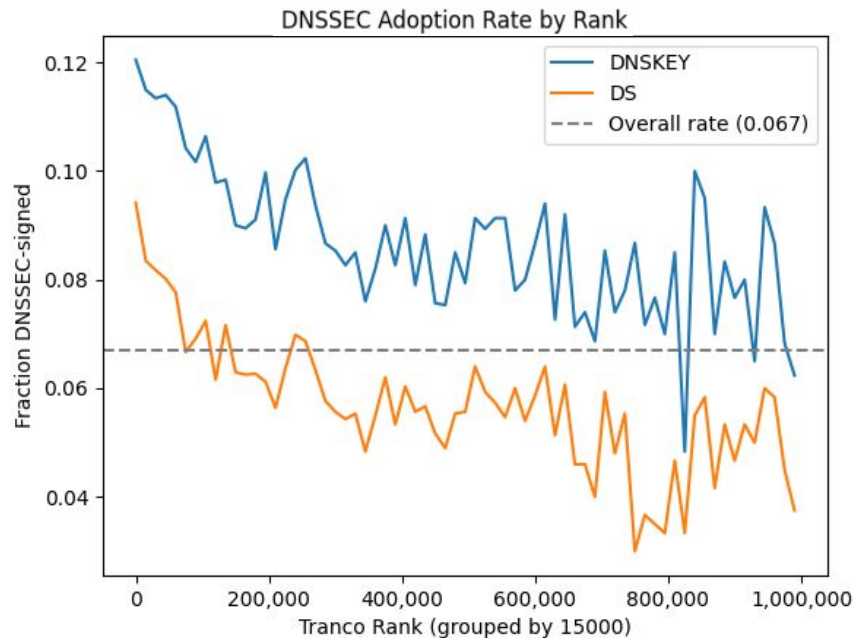
- **Signed:**
  - Domain returns non empty answers for DNSKEY and DS
- **Unsigned:**
  - Domain returns no answer for either DNSKEY or DS
- **Misconfigured:**
  - Domain returns a non empty answer for only one of the queries



Vast majority of domains are unsigned & most misconfigurations come from missing DS record

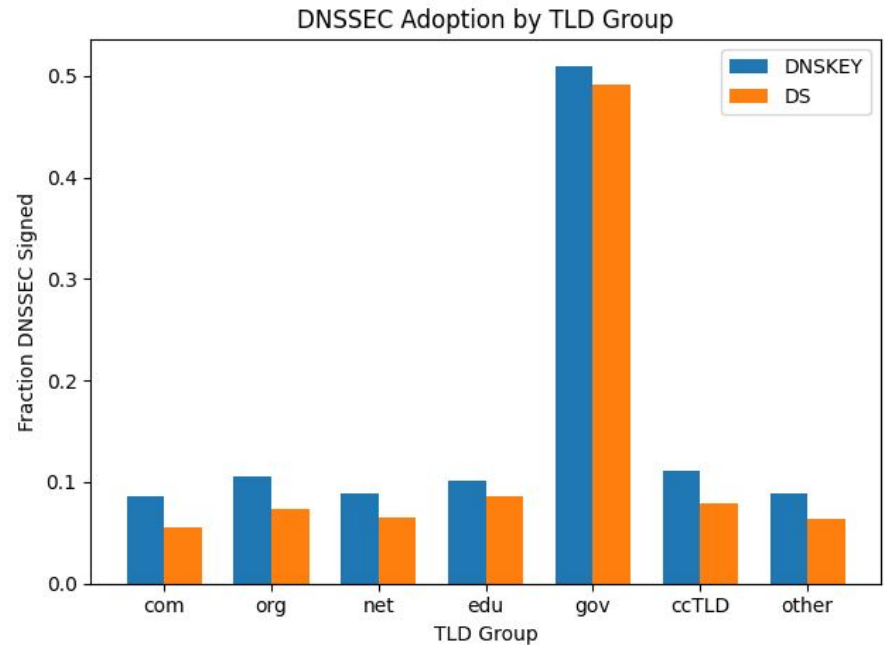
## RQ1: By Rank

- Signing rates are higher among top ranked domains, but drop off quickly



## RQ1: By TLD

- Signing rate is mostly uniform across TLDs , falling between 5-8%
- .gov domains are an outlier with a signing rate of 49%\*



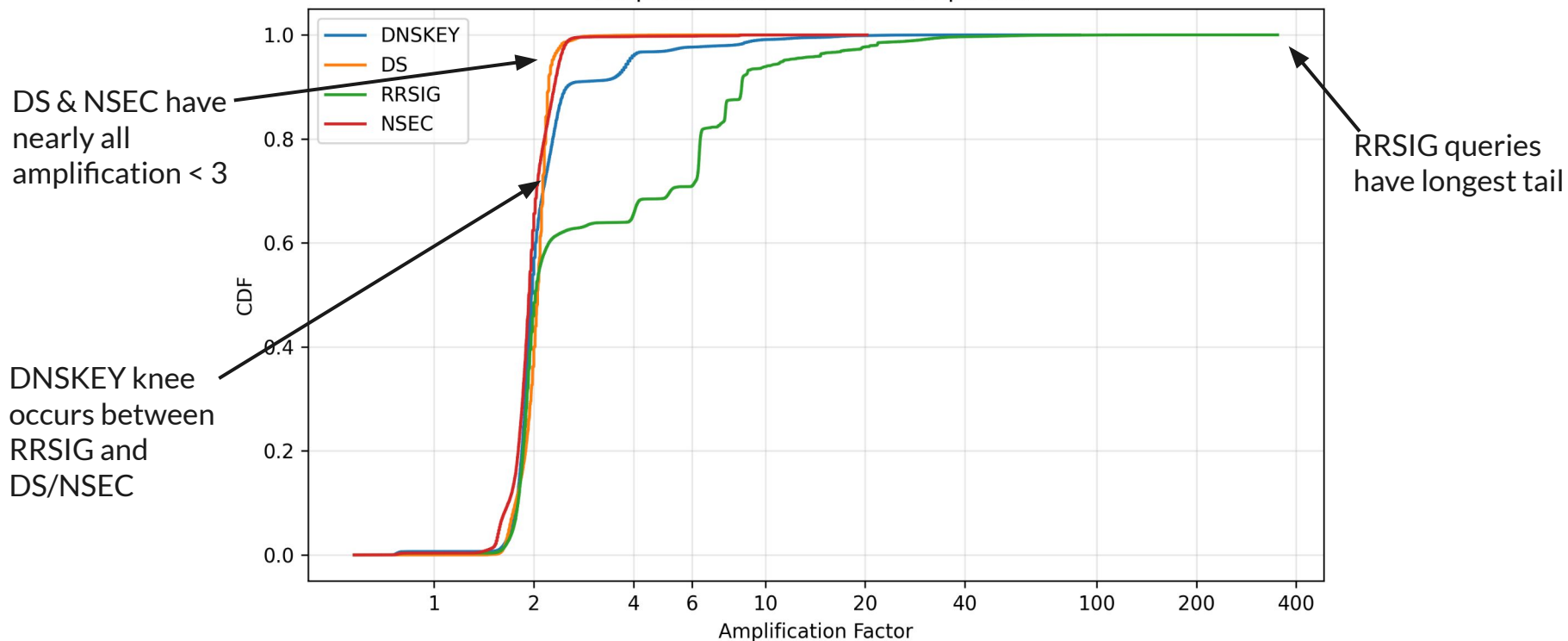
\*Reflects 2008 mandate of DNSSEC for federal domains



## RQ2/RQ3: Amplification Distribution

| Type   | Mean | Max    | P95   | P99   |
|--------|------|--------|-------|-------|
| DNSKEY | 2.33 | 88.83  | 3.90  | 9.58  |
| DS     | 2.03 | 11.25  | 3.27  | 2.62  |
| RRSIG  | 4.48 | 351.59 | 11.35 | 29.15 |
| NSEC   | 1.98 | 20.28  | 2.39  | 2.56  |
| All    | 2.7  | 351.59 | 7.16  | 15.10 |

Amplification Factor CDF Comparison



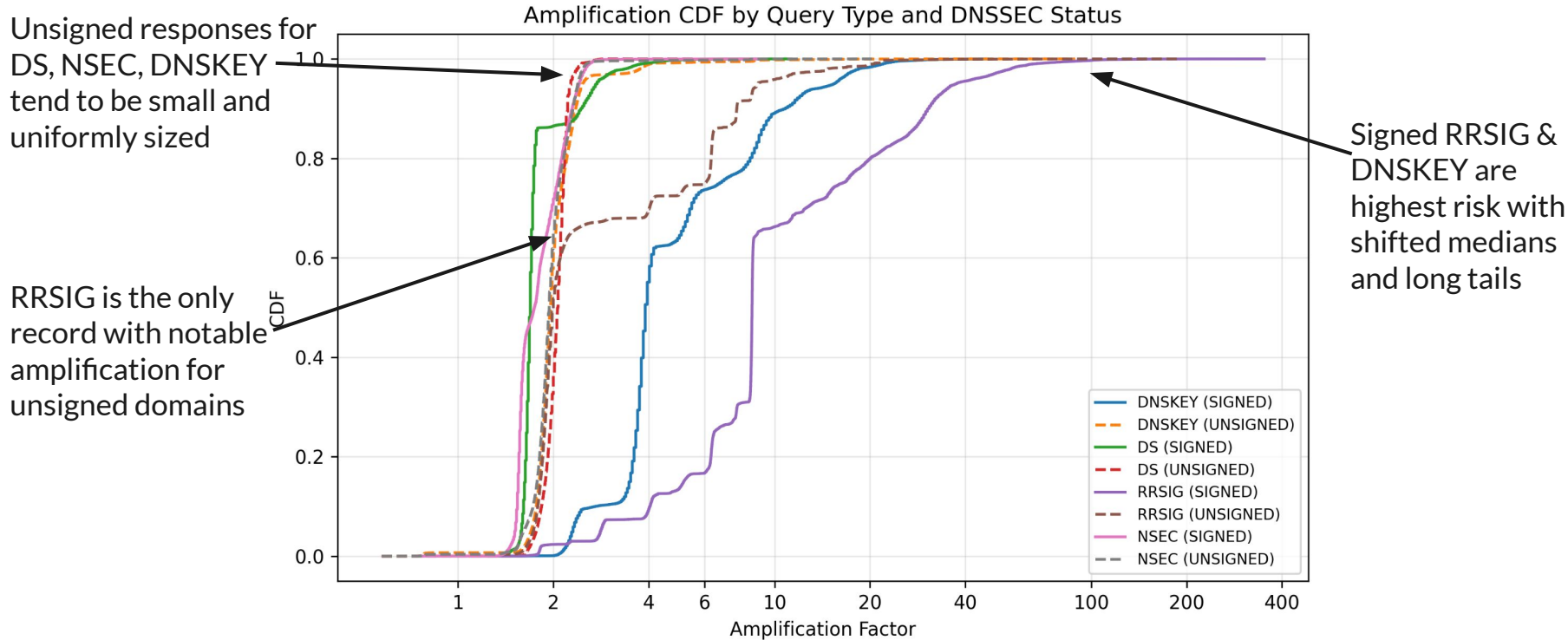
RRSIG queries have the highest variability in amplification and largest tail, while DS and NSEC queries tend to have more consistently low amplifications & DNSKEY sits between.



## RQ4: Signed vs Unsigned Behavior

- For all queries, signed domains produce higher amplification than unsigned
  - Signed typically return records while unsigned return denials

|      | Signed | Unsigned |
|------|--------|----------|
| Mean | 5.8    | 2.5      |
| Min  | 0.75   | 0.54     |
| Max  | 351.6  | 191.2    |
| P95  | 21.5   | 6.3      |
| P99  | 43.3   | 10.5     |



For the same query types, signed domains consistently produce larger responses than unsigned ones. The difference is especially pronounced in the tail ends, where the highest amplification risk is.



## RQ5: DNSSEC Properties

DNSKEY Amplification drivers:

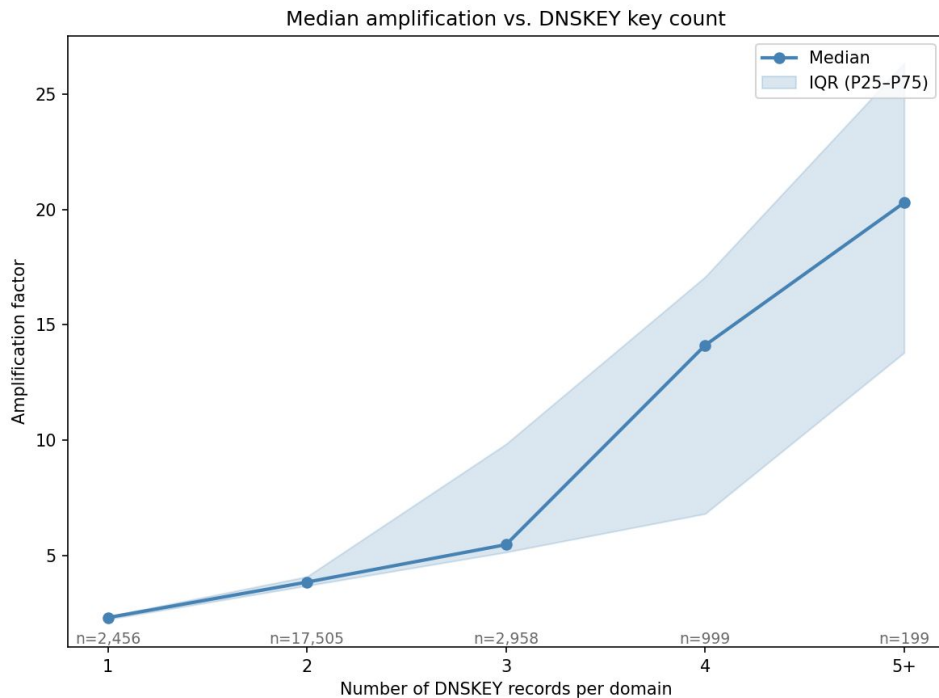
- Number of keys (ZSK, KSK)
- Key size (RSA vs EC)
- Encryption algorithm choice

RRSIG Amplification drivers:

- Number of signed records (RRSets)
- Record type coverage
- Encryption algorithm choice

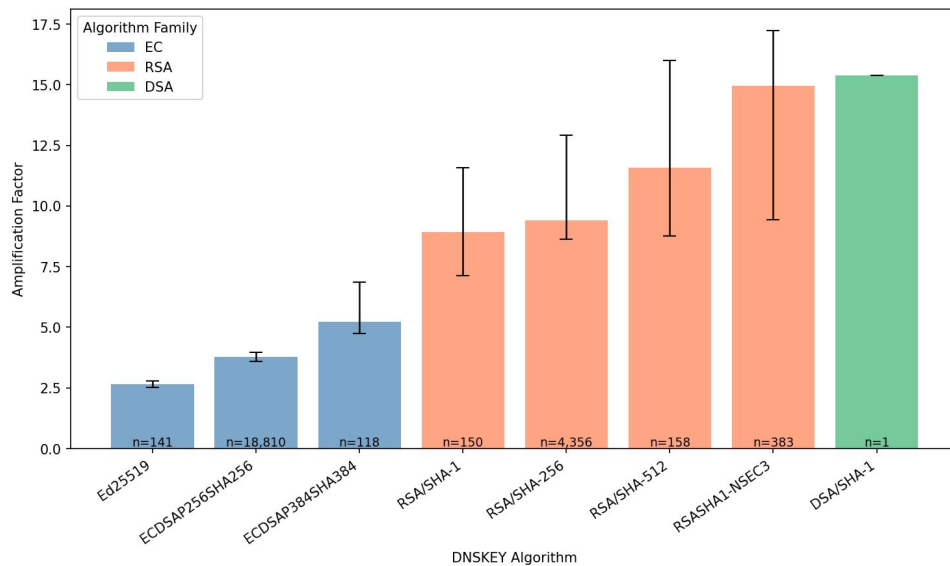
## RQ5: DNSKEY Key Count

- Amplification increases with key count, with a jump past 3 keys
- Variability in amplification also increases → more extreme cases
- Two keys is the most common configuration (ZSK & KSK)



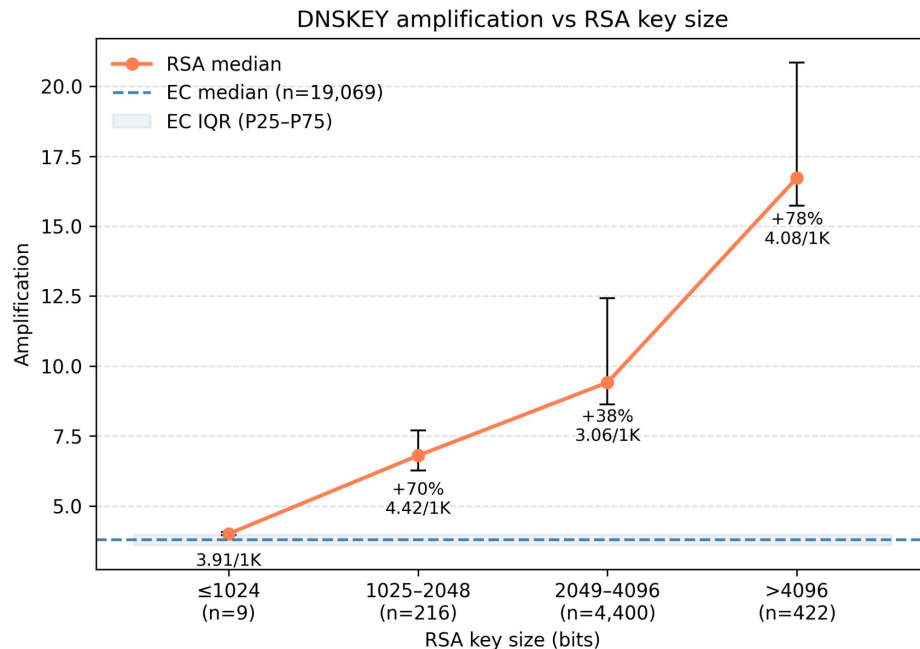
## RQ5: Encryption Algorithm

- EC has less amplification while RSA has higher amplification & variance
  - RSA amplification increases with stronger SHA hashes → larger key sizes
- ECDSA P-256 is most commonly used, reflecting current recommendations

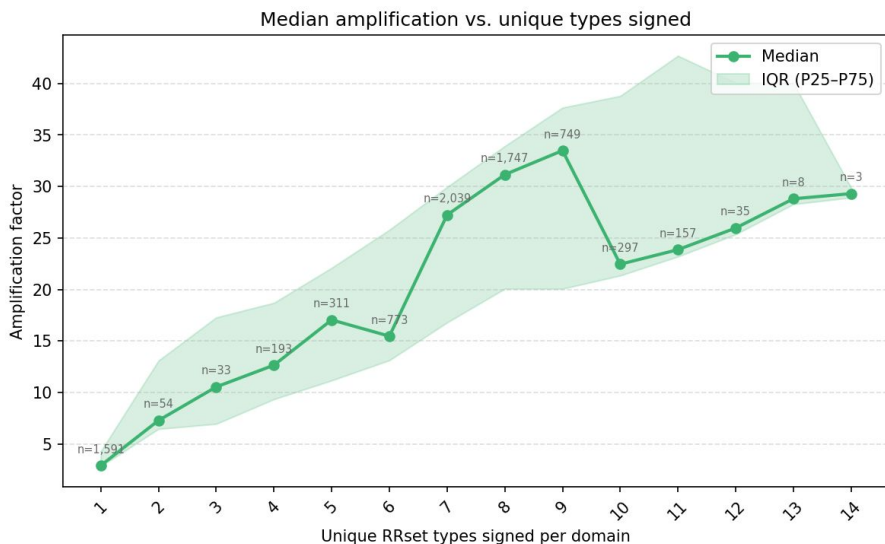


## RQ5: RSA Key Size

- Amplification increases with RSA key size
- Most common key size is between 2,049 - 4,096 bits
- Increase is somewhat linear, but with some jumps in places



## RQ5: RRSIG Record Coverage



We find some common RRSets:

|                                              | N    | Med  | P95  |
|----------------------------------------------|------|------|------|
| A, DNSKEY, MX, NS, NSEC3PARAM, SOA, TXT      | 1247 | 28.1 | 53.4 |
| A                                            | 853  | 2.9  | 6.7  |
| A, CDS, DNSKEY, MX, NS, NSEC3PARAM, SOA, TXT | 517  | 31.2 | 33.4 |
| A, DNSKEY, MX, NS, NSEC, SOA, TXT            | 394  | 17.6 | 52.5 |



# Conclusion

## Findings:

- DNSSEC adoption is low (6.7%)
  - Concentrated in high-ranked and .gov domains
- Signed domains produce higher amplification — especially from RRSIG and DNSKEY queries
- Amplification is driven by configuration choices (key count, RSA vs EC, and zone size)

## Future areas:

- Track how amplification changes as DNSSEC adoption and algorithms evolve over time